

MNYMO Estimathon 2025

MINNESOTA YOUTH MATH OUTREACH

August 30th, 2025

Problems for the MNYMO Estimathon 2025. As a reminder, the score for your team is

$$\left(10 + \sum_{\text{good intervals}} \left\lfloor \frac{\max}{\min} \right\rfloor\right) \cdot 2^{13-(\# \text{ of good intervals})}.$$

All answers are positive reals.

Problem 1. Let a and b represent the two shortest distances (in miles) from St. Paul Academy to Interstate highways whose route numbers are perfect squares. Estimate the product ab .

Problem 2. For letters ℓ , let $f(\ell)$ be the frequency at which ℓ appears in the English language, uppercase and lowercase, according to Wikipedia. Estimate

$$\frac{1}{f(m)^2} + \frac{1}{f(n)^2} + \frac{1}{f(y)^2} + \frac{1}{f(m)^2} + \frac{1}{f(o)^2}.$$

Problem 3. Estimate the number of characters, excluding spaces and comments, in the L^AT_EX file of the 2025 MNYMO General Round.

Problem 4. Recently, a new Superman movie directed by James Gunn was released and earned around 500 million dollars at the box office. Estimate the number of dollars that the first creators of Superman initially sold him for.

Problem 5. Estimate the total weight, in pounds, of all the McDonald's fries served worldwide in a day.

Problem 6. Estimate the length of the official *The Summer I Turned Pretty* playlist, as of Season 3 Episode 7, in seconds.

Problem 7. Estimate the number of references on the Wikipedia page for *Wikipedia*, as of August 21st, 2025.

Problem 8. Estimate the average number of Spotify streams for each solo Drake song, as of August 21st, 2025.

Problem 9. Estimate the number of frames throughout the entirety of the Netflix film *K-Pop Demon Hunters*.

Problem 10. Angie has a workbook with 20 problems, numbered from 1 through 20. She uses a spinner to determine which problem she should do next, with each problem from 1 – 20 being equally likely. If the spinner lands on a problem Angie has already done, she spins it again. Angie is done when she spins the spinner 3 times in a row without doing any new problems. The expected number of problems until she is done is $\frac{p}{q}$, where p and q are relatively prime positive integers. Estimate $p + q$.

Problem 11. Estimate the second-largest number of official AoPS posts out of all AoPS users, as of August 21st, 2025.

Problem 12. Estimate the value of the product of the number of Tom Brady's career touchdowns and Michael Jordan's career three-pointers.

Problem 13. Estimate the value of the product of the three least answers out of all previous problems.